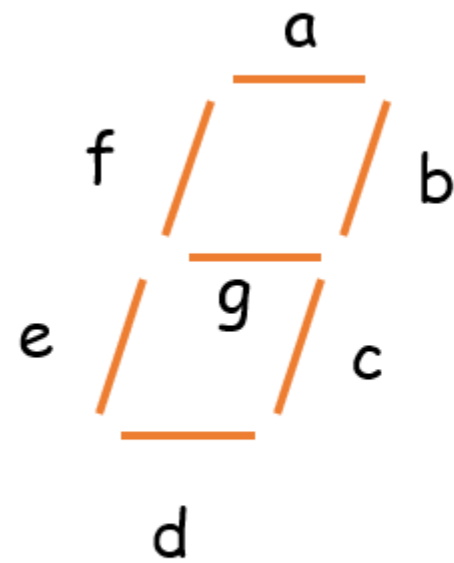


7-Segment Decoder



Assumption: Input represents a legal digit (0-9)

CD \ AB	00	01	11	10
00	1		1	1
01		1	1	1
11	X	X	X	X
10	1	1	X	X

$$A + C + BD + B'D'$$

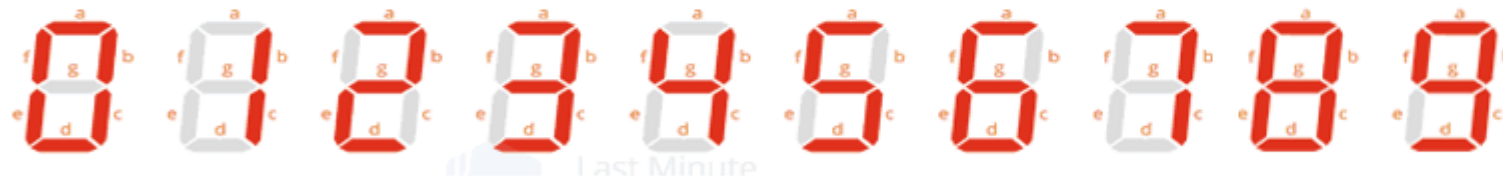
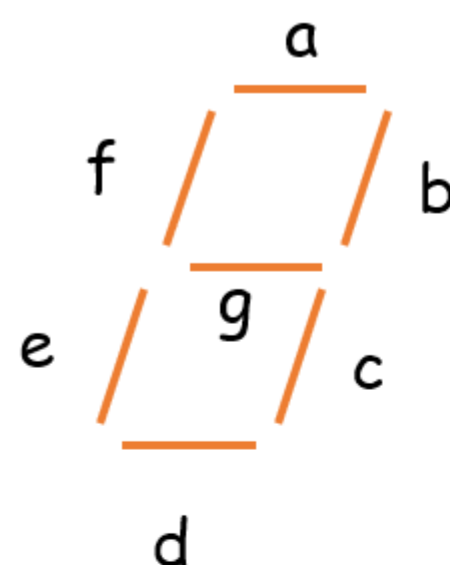


Table for a

	A	B	C	D	a
0	0	0	0	0	1
1	0	0	0	1	0
2	0	0	1	0	1
3	0	0	1	1	1
4	0	1	0	0	0
5	0	1	0	1	1
6	0	1	1	0	1
7	0	1	1	1	1
8	1	0	0	0	1
9	1	0	0	1	1
10					X
11					X
12					X
13					X
14					X
15					X

7-Segment Decoder



Assumption: Input represents a legal digit (0-9)

CD \ AB	00	01	11	10
00	/	/	/	/
01	/	0	/	0
11	X	X	X	X
10	/	/	X	X

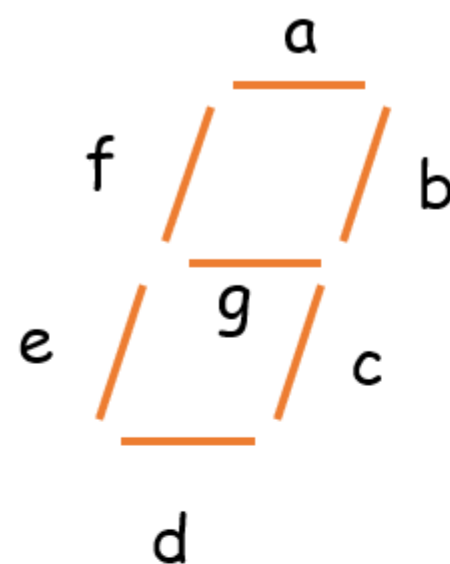
$$B' + CD + C'D'$$



Table for b

	A	B	C	D	b
0	0	0	0	0	/
1	0	0	0	1	/
2	0	0	1	0	1
3	0	0	1	1	1
4	0	1	0	0	/
5	0	1	0	1	0
6	0	1	1	0	G
7	0	1	1	1	1
8	1	0	0	0	1
9	1	0	0	1	1
10					X
11					X
12					X
13					X
14					X
15					X

7-Segment Decoder



Assumption: Input represents a legal digit (0-9)

CD	00	01	11	10
AB				
00	/	/	/	0
01	/	/	/	/
11	X	X	X	X
10	/	/	X	X

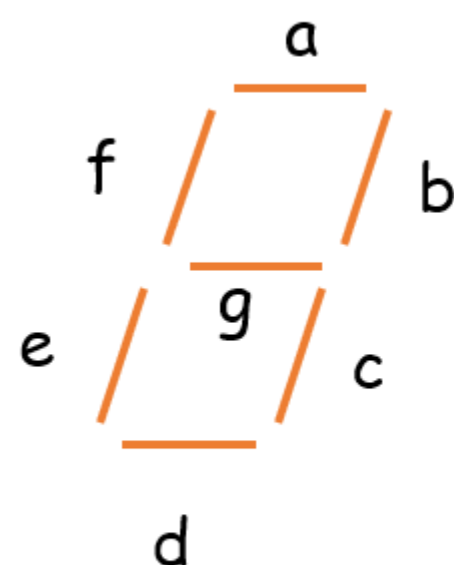
$$B + C' + D'$$



Table for c

	A	B	C	D	c
0	0	0	0	0	/
1	0	0	0	1	/
2	0	0	1	0	0
3	0	0	1	1	/
4	0	1	0	0	/
5	0	1	0	1	/
6	0	1	1	0	/
7	0	1	1	1	/
8	1	0	0	0	/
9	1	0	0	1	/
10					X
11					X
12					X
13					X
14					X
15					X

7-Segment Decoder



Assumption: Input represents a legal digit (0-9)

CD \ AB	00	01	11	10
00	/	0	/	/
01	0	/	0	/
11	X	X	X	X
10	/	/	X	X

$$A + B'D' + B'C + CD'$$

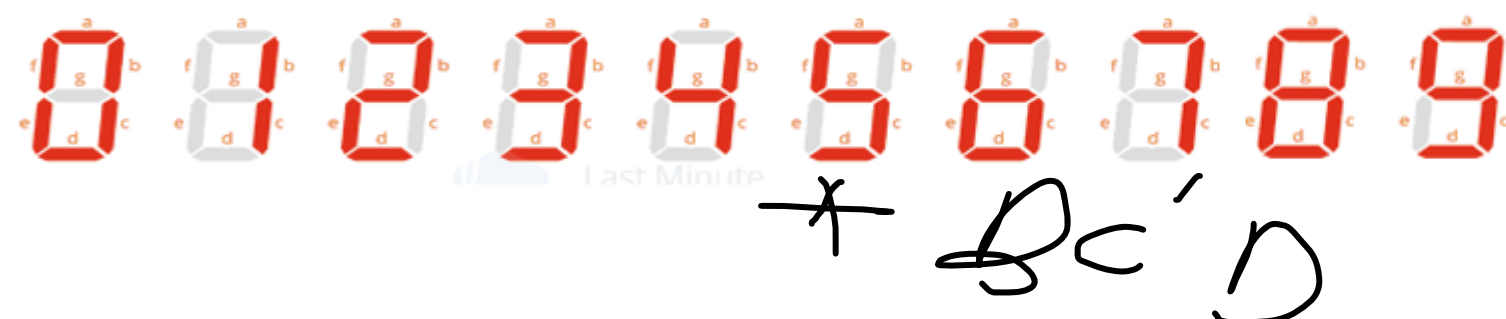
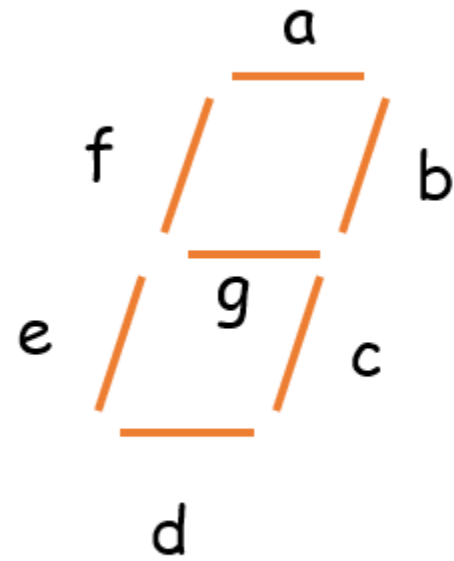


Table for d

	A	B	C	D	d
0	0	0	0	0	/
1	0	0	0	1	0
2	0	0	1	0	/
3	0	0	1	1	/
4	0	1	0	0	0
5	0	1	0	1	/
6	0	1	1	0	/
7	0	1	1	1	0
8	1	0	0	0	/
9	1	0	0	1	/
10					X
11					X
12					X
13					X
14					X
15					X

7-Segment Decoder



Assumption: Input represents a legal digit (0-9)

CD \ AB	00	01	11	10
00	1			1
01				1
11	X	X	X	X
10	1		X	X

$$CD' + B'D'$$

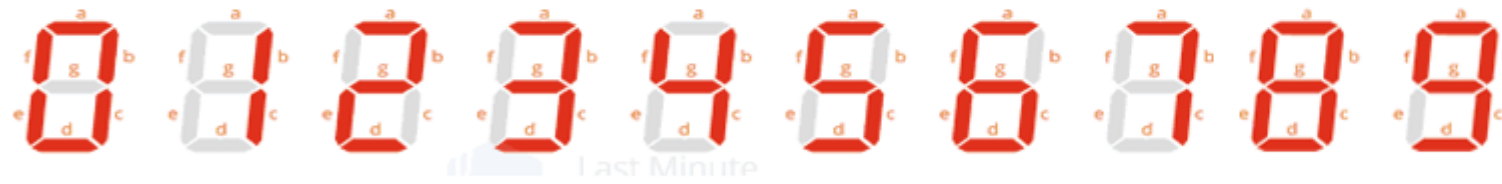
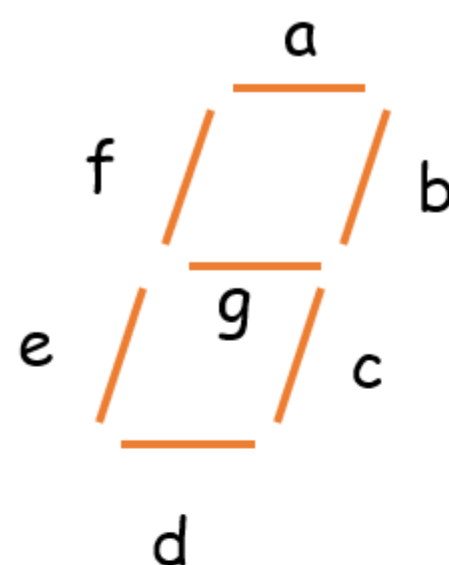


Table for e

	A	B	C	D	e
0	0	0	0	0	1
1	0	0	0	1	0
2	0	0	1	0	1
3	0	0	1	1	0
4	0	1	0	0	0
5	0	1	0	1	0
6	0	1	1	0	1
7	0	1	1	1	0
8	1	0	0	0	1
9	1	0	0	1	0
10					X
11					X
12					X
13					X
14					X
15					X

7-Segment Decoder



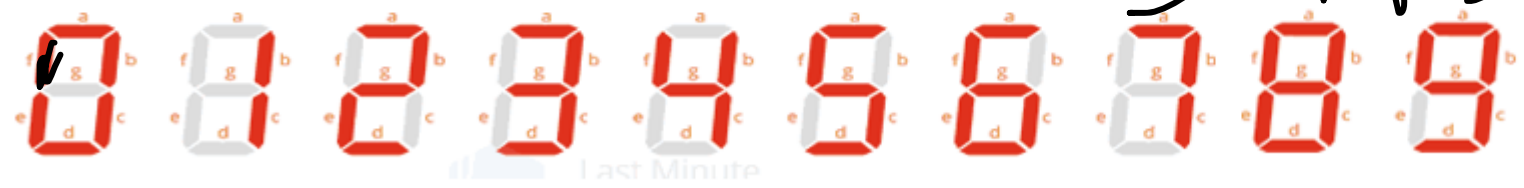
Assumption: Input represents a legal digit (0-9)

CD \ AB	00	01	11	10
00				
01				
11				
10				

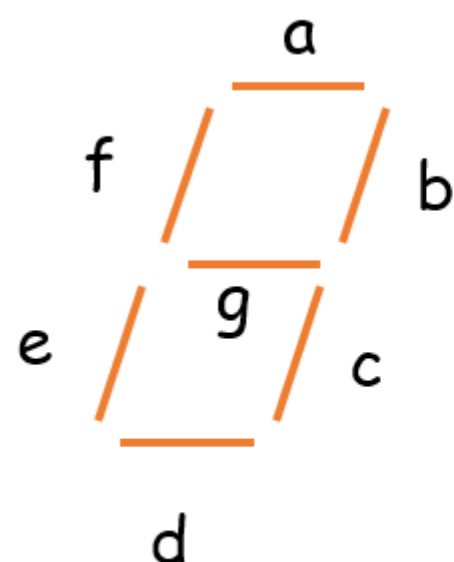
Table for f

	A	B	C	D	f
0	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	2
3	0	0	1	1	3
4	0	1	0	0	4
5	0	1	0	1	5
6	0	1	1	0	6
7	0	1	1	1	7
8	1	0	0	0	8
9	1	0	0	1	9
10					X
11					X
12					X
13					X
14					X
15					X

$$A + B C' + C' D + B D$$



7-Segment Decoder



Assumption: Input represents a legal digit (0-9)

PC

CD	00	01	11	10
AB				
00	0	0	1	1
01	1	1	0	1
11	X	X	X	X
10	1	1	X	X

Table for g

	A	B	C	D	b
0	0	0	0	0	0
1	0	0	0	1	0
2	0	0	1	0	1
3	0	0	1	1	1
4	0	1	0	0	1
5	0	1	0	1	1
6	0	1	1	0	1
7	0	1	1	1	0
8	1	0	0	0	1
9	1	0	0	1	1
10					X
11					X
12					X
13					X
14					X
15					X

